3RD WEEK OF DEVELOPMENT II

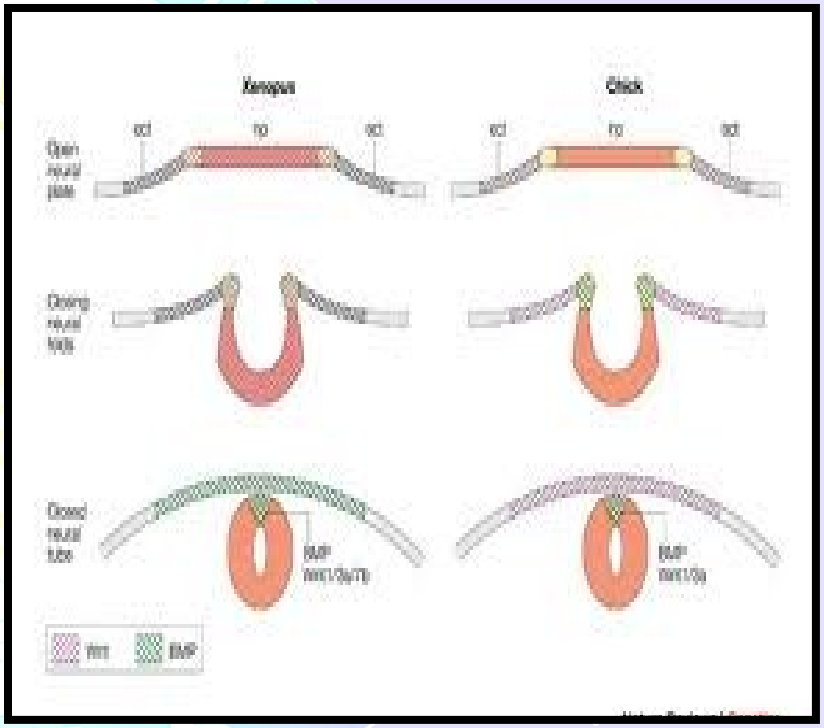
NERULATION & DEVELOPMENT OF SOMITES

LEARNING OBJECTIVES

- At the end of the lecture student should be able to:
  - Know what do mean by **neurulation.**
  - Explain development of the **Intraembryonic coelom.**
  - Describe the development of **chorionic villi.**

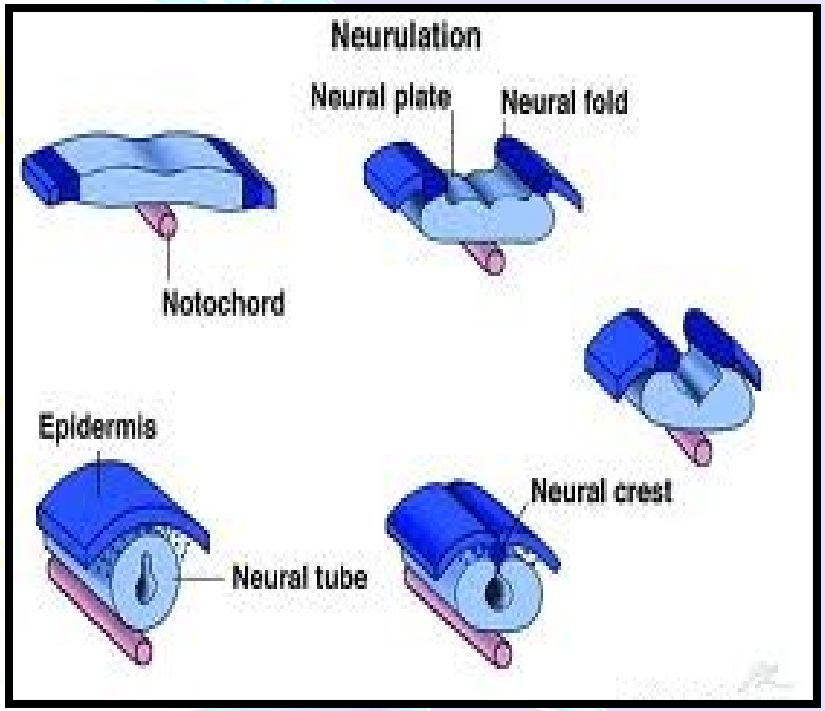
DEFINITION OF NEURULATION

- The development of neural tube is called as the Neurulation. This process involves formation of
  - Neural plate
  - Neural folds
  - Closure of the folds
  - Neural tube
- This process is completed by the end of the fourth week, when closure of the caudal **neuropore** occurs.
- During neurulation, the embryo may be referred to as a **neurula**.



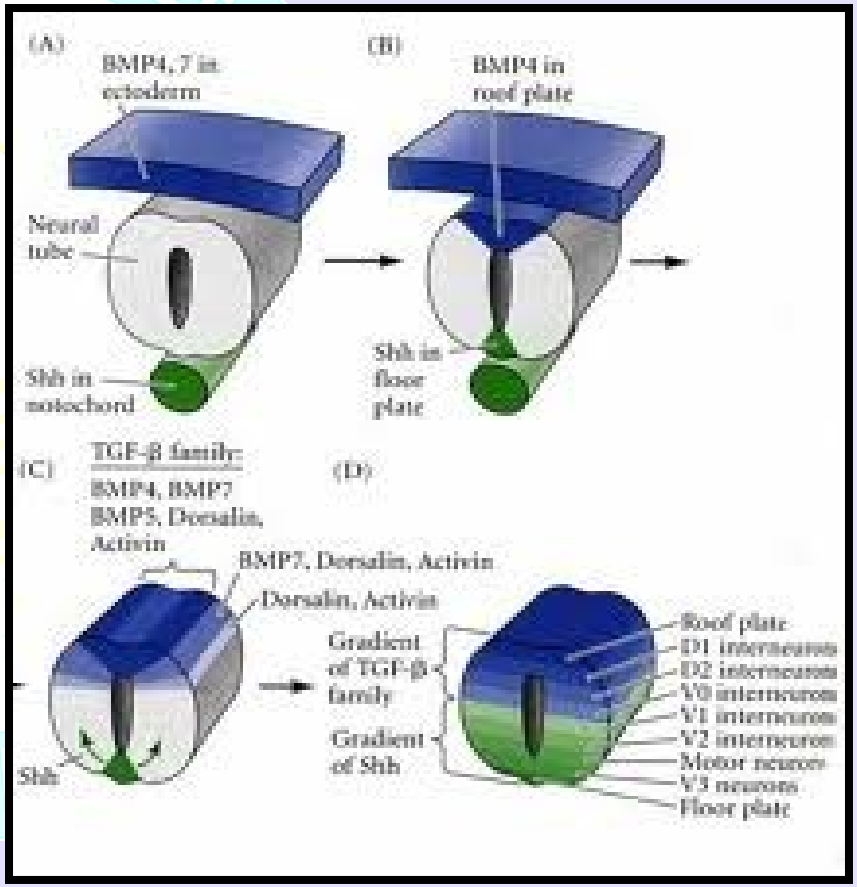
DEVELOPMENT OF NEURAL PLATE

- As the notochord develops, it induces the overlying embryonic ectoderm located at or adjacent to the midline to thicken and form an elongated plate of thickened epithelial cells, the **neural plate**.
- The ectoderm of the neural plate (neuroectoderm) gives rise to the brain and spinal cord.



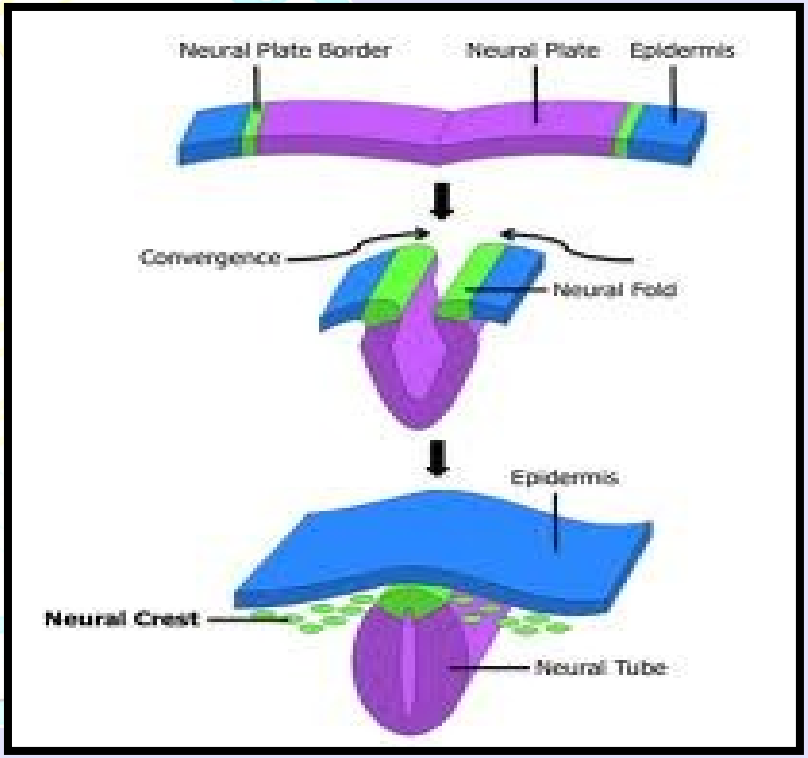
DEVELOPMENT OF NEURAL PLATE CONT..

- The elongated neural plate corresponds in length to the underlying notochord.
- It appears rostral to the primitive node and dorsal to the notochord and the mesoderm adjacent to it.
- As the notochord elongates, the neural plate broadens and eventually extends cranially as far as the oropharyngeal membrane.
- Eventually the neural plate extends beyond the notochord.



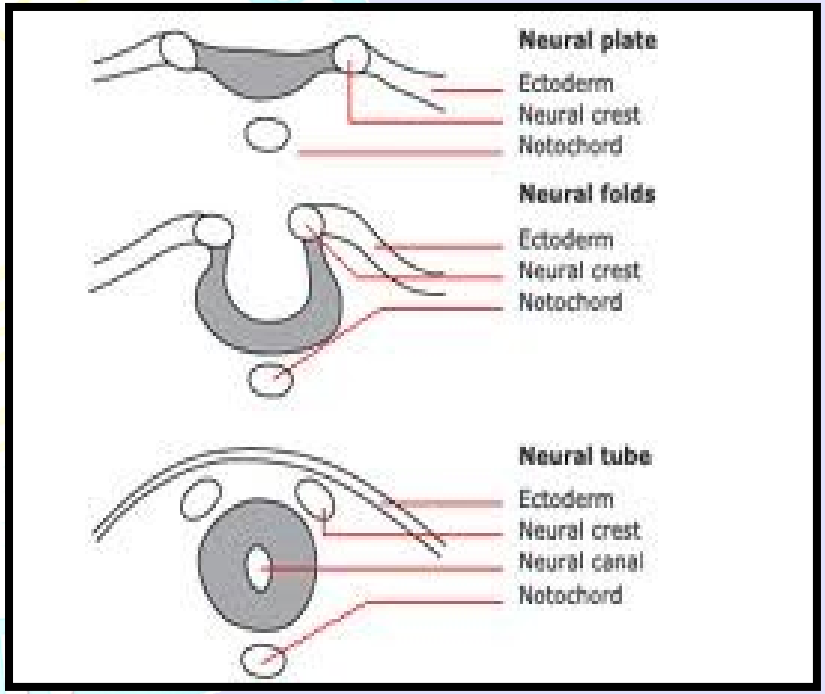
NEURAL GROOVE AND FOLDS

- On approximately the 18th day, the neural plate invaginates along its central axis to form a longitudinal median **neural groove**, which has neural folds on each side.
- The **neural folds** become particularly prominent at the cranial end of the embryo and are the first signs of brain development.



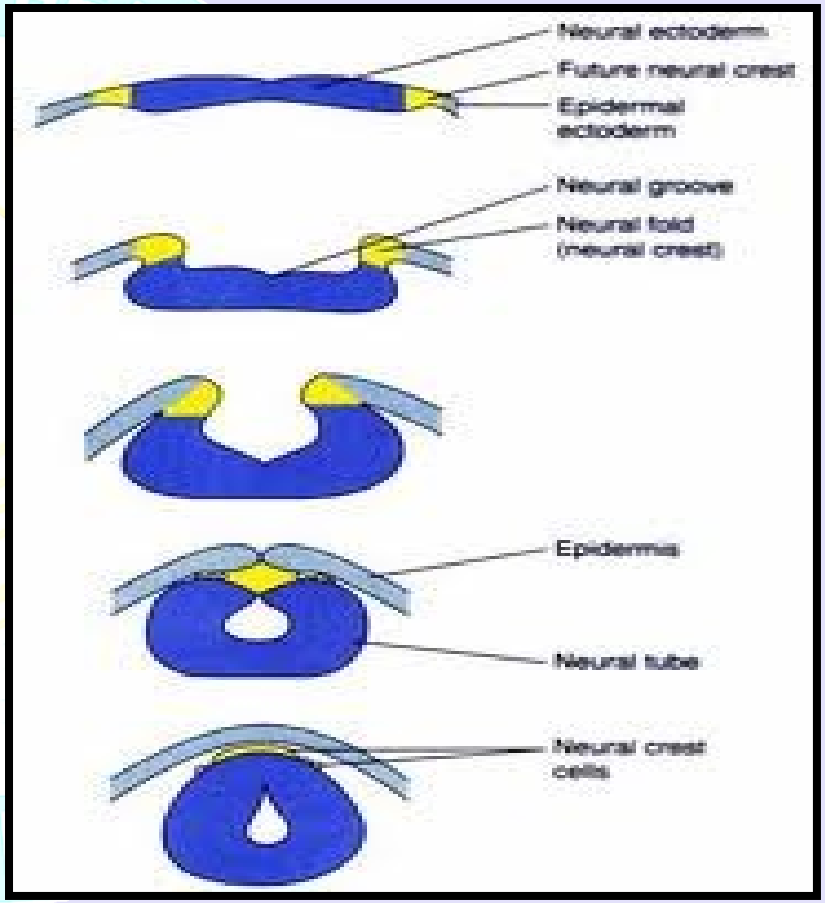
NEURAL TUBE

- By the end of the third week, the neural folds have begun to move together and fuse, converting the neural plate into a **neural tube**, the primordium of the CNS
- The neural tube soon separates from the surface ectoderm as the neural folds meet



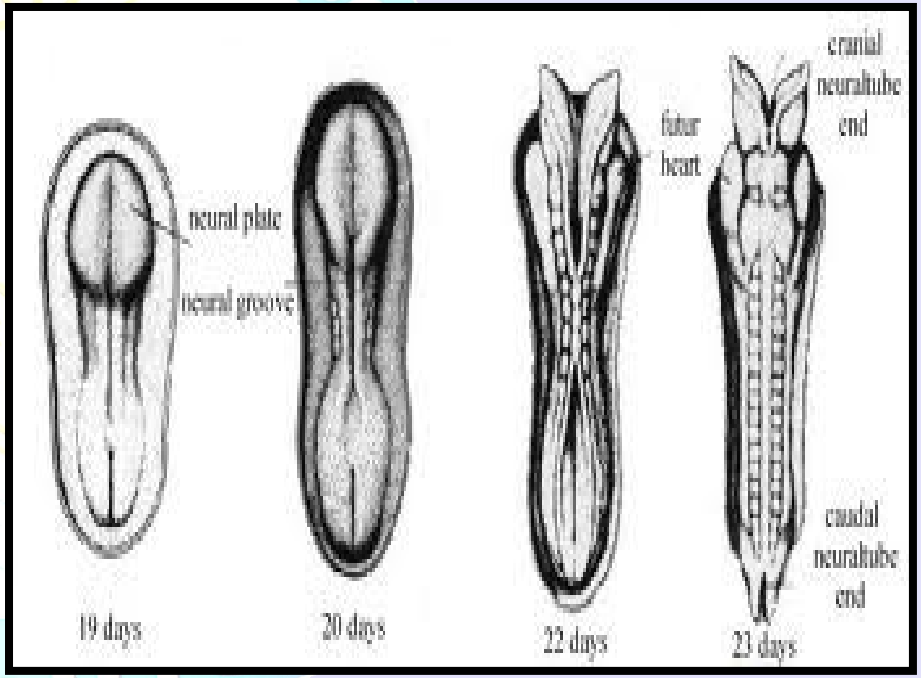
SEPARATION OF NEURAL TUBE FROM SURFACE ECTODERM

- Neural crest cells undergo an epithelial to mesenchymal transition and migrate away as the neural folds meet
- the free edges of the surface ectoderm (non-neural ectoderm) fuse so that this layer becomes continuous over the neural tube and the back of the embryo.
- The surface ectoderm differentiates into the epidermis.



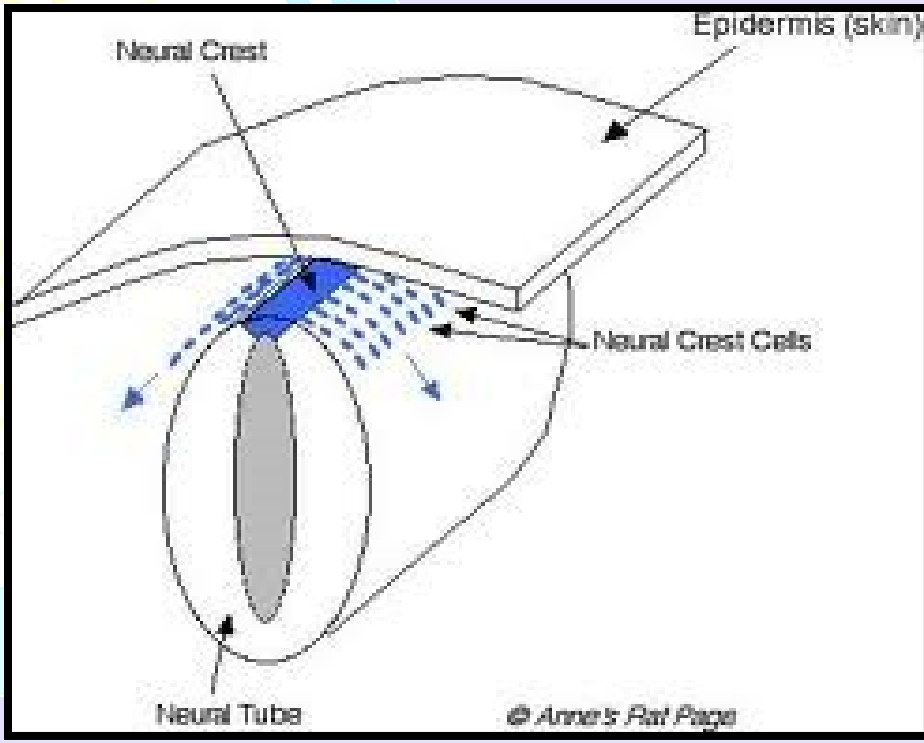
COMPLETION OF NEURAL TUBE

- Neurulation is completed during the fourth week.
- Neural tube formation is a complex cellular and multifactorial process involving a cascade of *molecular mechanisms* and extrinsic factors.



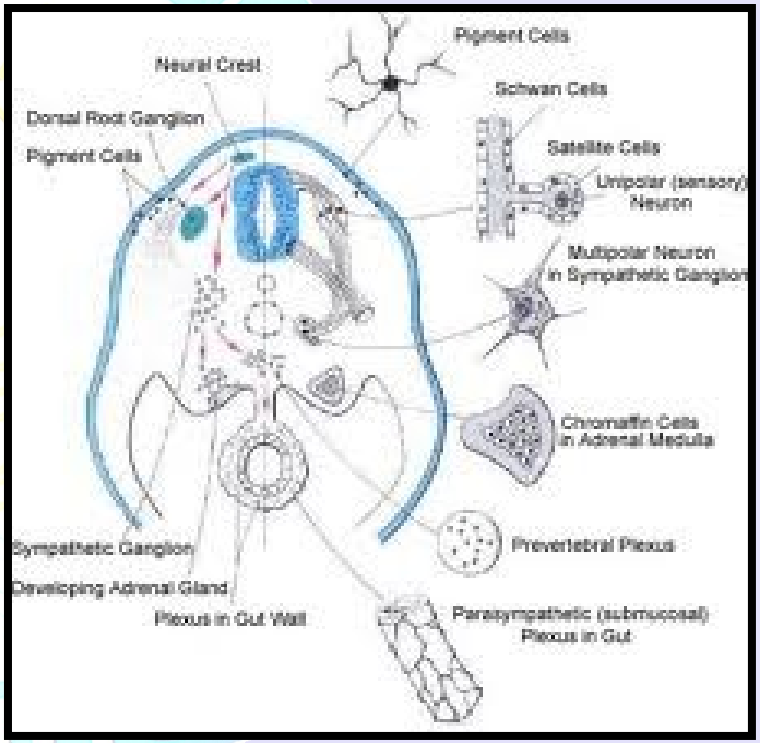
NEURAL CREST

- As the neural folds fuse to form the neural tube, some neuroectodermal cells lying along the inner margin of each neural fold lose their epithelial affinities and attachments to neighboring cells.
- As the neural tube separates from the surface ectoderm, **neural crest cells** form a flattened irregular mass, the **neural crest**, between the neural tube and the overlying surface ectoderm



DERIVATIVES OF NEURAL CREST

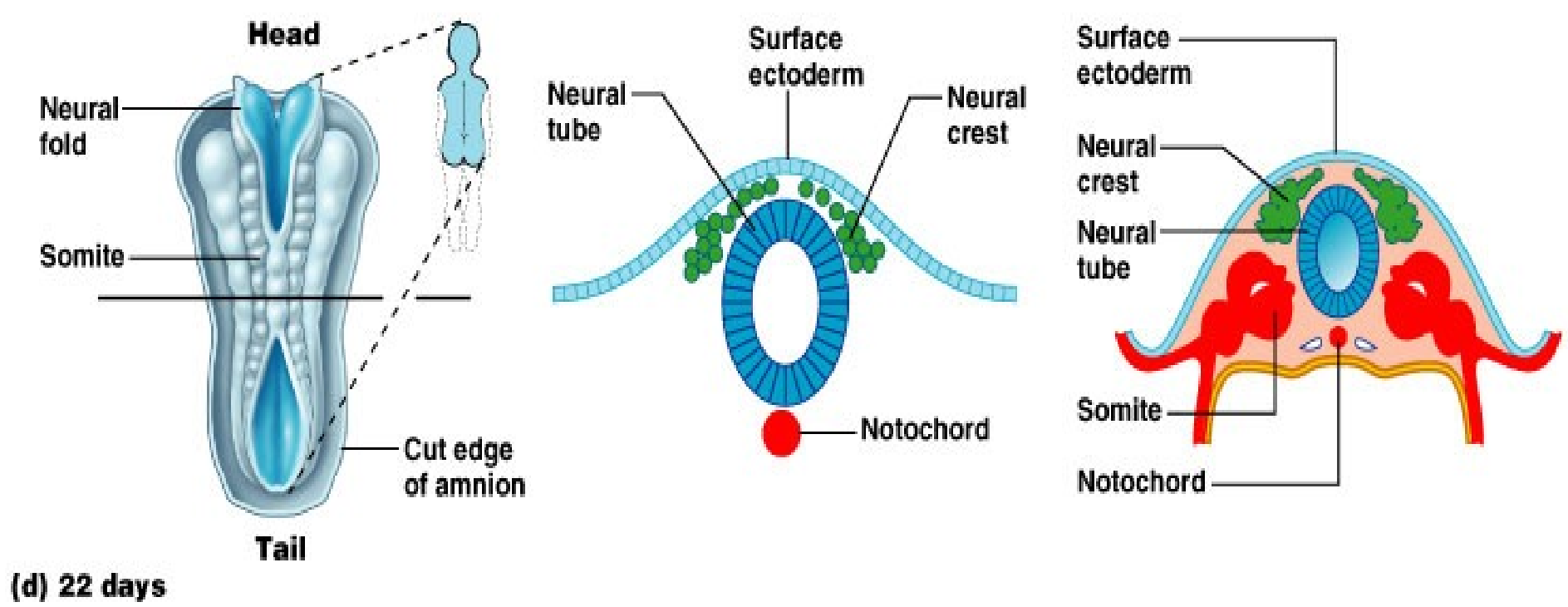
- The neural crest soon separates into right and left parts that shift to the dorsolateral aspects of the neural tube; here they give rise to the sensory ganglia of the spinal and cranial nerves.
- Neural crest cells give rise to the:
  - spinal ganglia (dorsal root ganglia)
  - ganglia of the autonomic nervous system.
  - The ganglia of cranial nerves V, VII, IX, and X are also partly derived from neural crest cells.
  - the neurolemma sheaths of peripheral nerves
  - the formation of the leptomeninges
  - The pigment cells
  - the suprarenal (adrenal) medulla



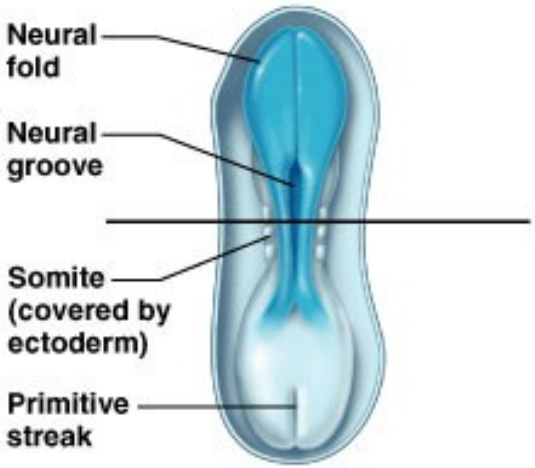
NEURAL TUBE DEFECTS

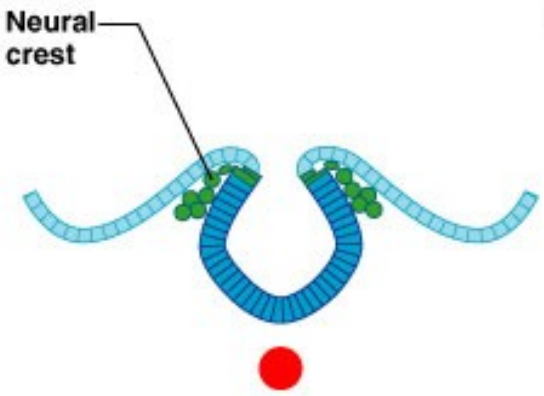
- Among the most common congenital anomalies,
- **Meroencephaly** (partial absence of the brain) is the most severe neural tube defect and is also the most common anomaly affecting the CNS.
- Although the term anencephaly (Gr. *an*, without + *enkephalos*, brain) is commonly used, it is a misnomer because a remnant of the brain is present.
- Available evidence suggests that the primary disturbance (e.g., a teratogenic drug 20) affects cell fates, cell adhesion, and the mechanism of neural tube closure.
- This results in failure of the neural folds to fuse and form the neural tube.

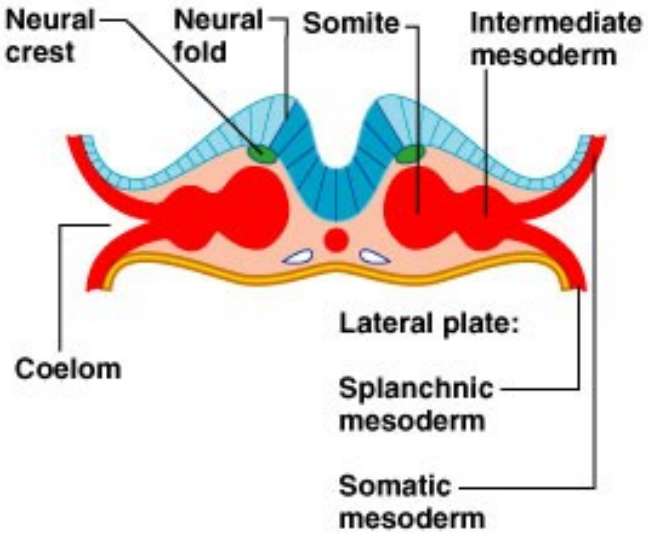




- Closure of neural tube: begins at end of week 3; complete by end of week 4 (folic acid important for this step)
- Extends cranially (eventually brain) and caudally (spinal cord)
- Neural crest, lateral ectodermal cells, pulled along and form sensory nerve cells and other structures

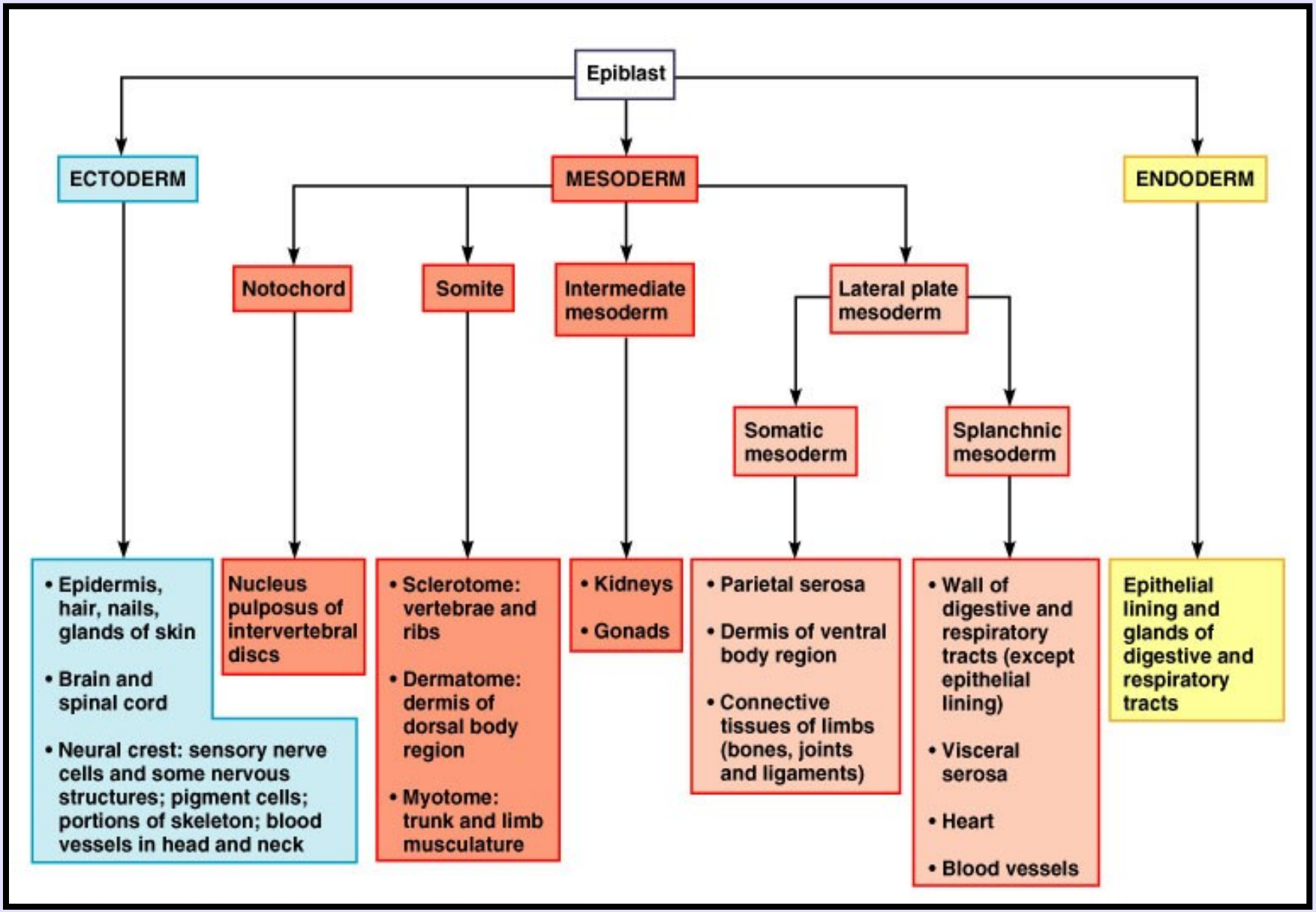






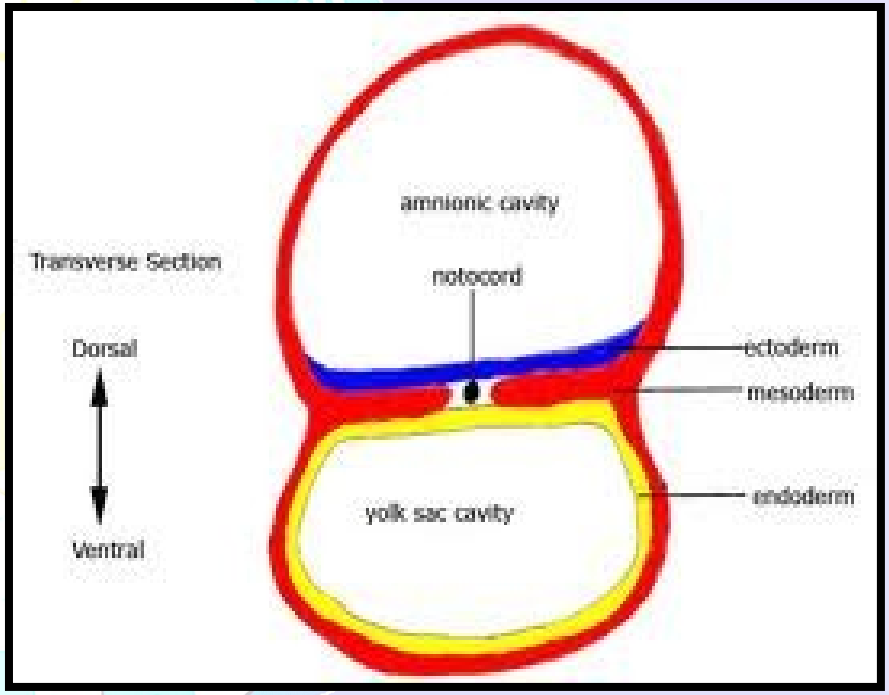
- Mesoderm begins to differentiate
  - Lateral to notochord, week 3
  - Extends cranially and caudally (from head to tail or crown to rump)
- Division of mesoderm into three regions
  - Somites: 40 pairs of body segments (repeating units, like building blocks) by end week 4
  - Intermediate mesoderm: just lateral to somites
  - Lateral plate: splits to form coelom (“cavity”)

Major Derivatives of the Embryonic Germ Layers



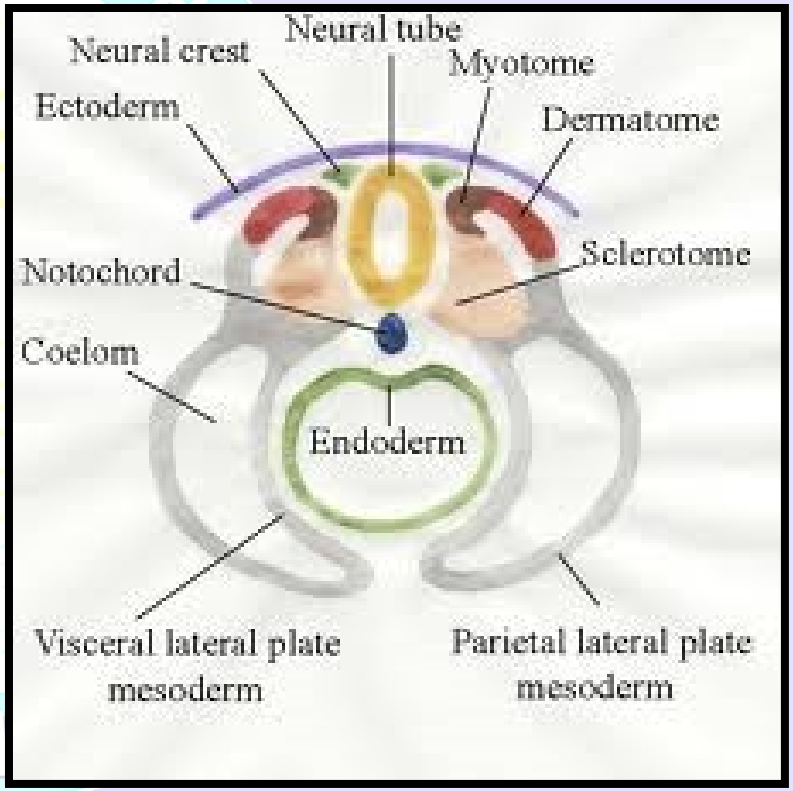
DEVELOPMENT OF MESODERM

- Mesoderm develops from the notochord and primitive streak
- Cells from the notochord and primitive streak they migrate rosteral and laterally and collected between two layer of epiblast (ectoderm) and hypoblast (endoderm)
- In this away three germ layers will develop



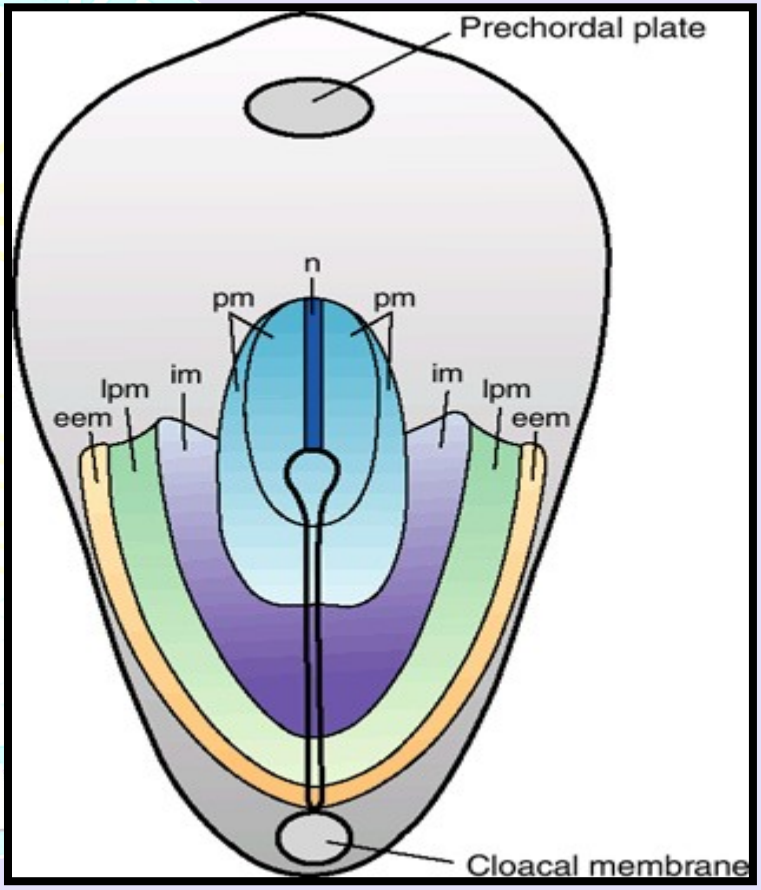
DIVISION OF MESODERM

- Embryologically the Mesoderm divided into three components each derived from separate source
- The notochord, cells derived from the primitive node form **parial maxesoderm**.
- Still more lateral cells from the primitive streak will form the longitudinal columns called as **intermediate mesoderm**, which gradually thins into a layer of lateral mesoderm.



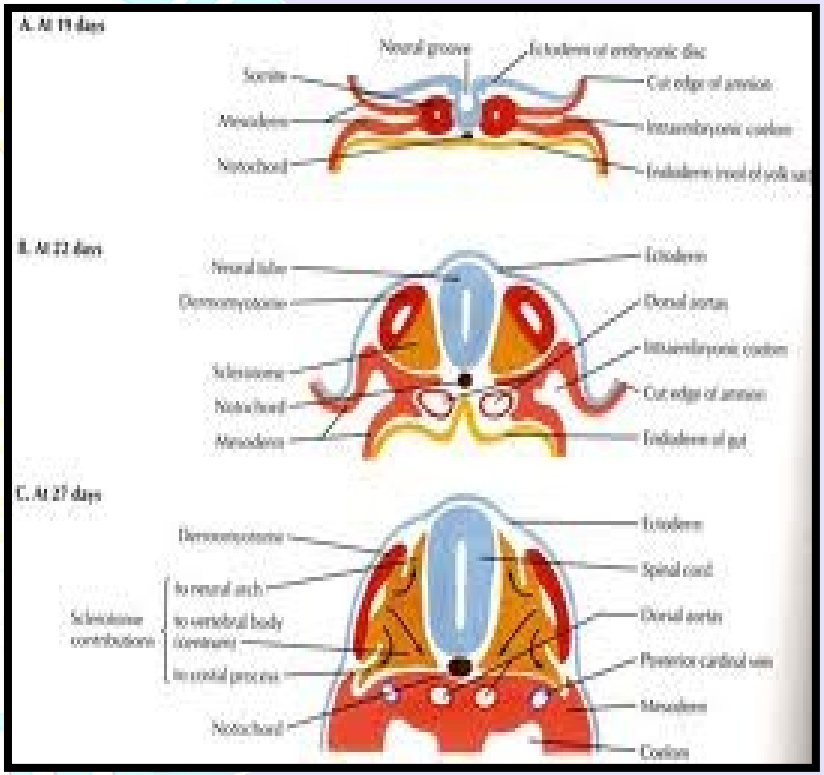
DIVISION OF MESODERM

- Close to the node, this population appears as a thick, longitudinal column of cells .
- Because the somites are so prominent during the fourth and fifth weeks, they are used as one of several criteria for determining an embryo's age.



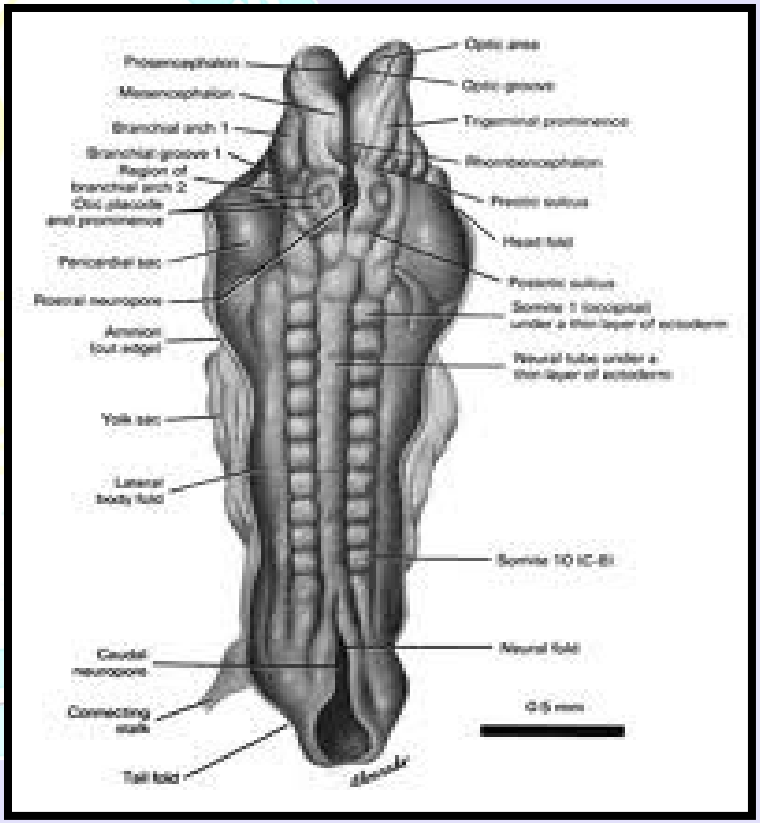
DIVISION OF MESODERM

- Still lateral the cells from primitive streak develop into the **lateral mesoderm** is continuous with the extraembryonic mesoderm covering the umbilical vesicle and amnion.



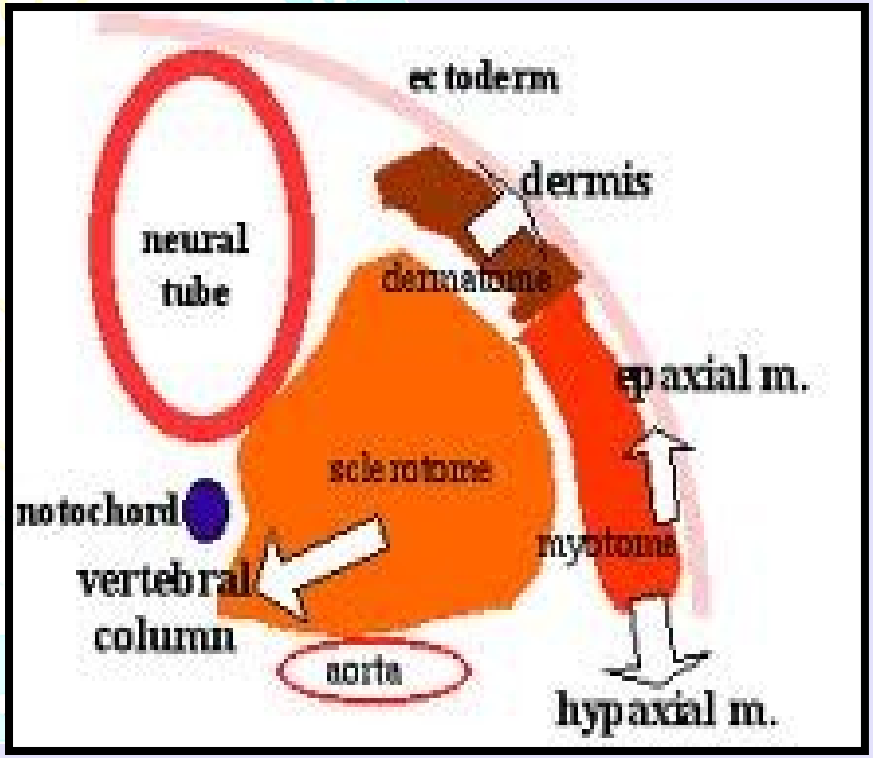
DEVELOPMENT OF SOMITES

- Toward the end of the third week, the **paraxial mesoderm** differentiates, condenses, and begins to divide into paired cuboidal bodies, the **somites** (Gr. *soma*, body)
- which form in a craniocaudal sequence.
- These blocks of mesoderm are located on each side of the developing neural tube.



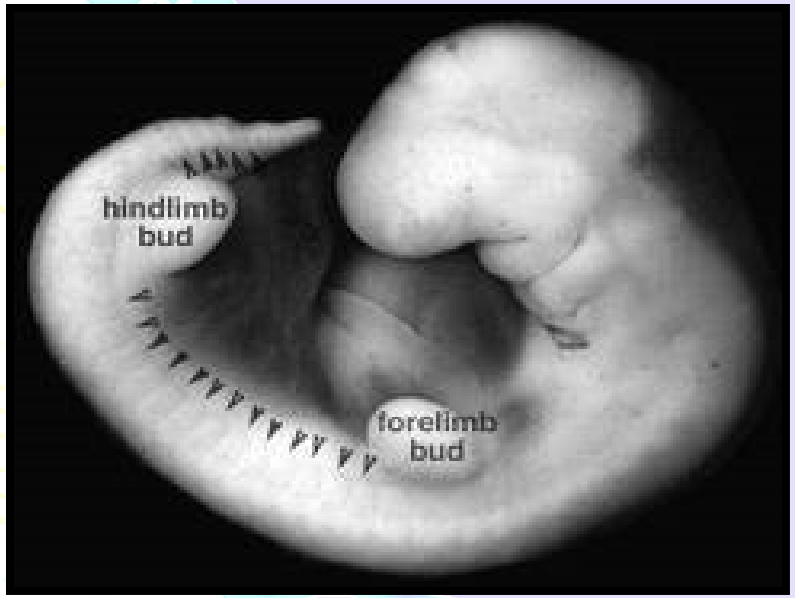
DEVELOPMENT OF SOMITES

- About 38 pairs of somites form during the *somite period of human development* (days 20 to 30).
- By the end of the fifth week, 42 to 44 pairs of somites are present.
- The somites form distinct surface elevations on the embryo and are somewhat triangular in transverse section.



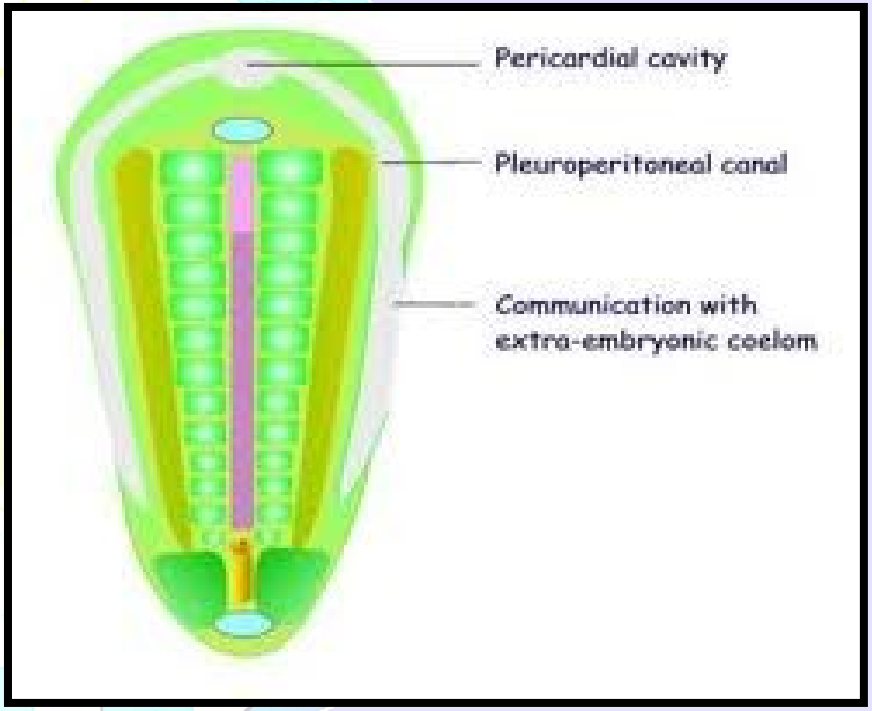
SOMITE

- **Somites** first appear in the future occipital region of the embryo.
- They soon develop craniocaudally and give rise to most of the *axial skeleton* and associated musculature as well as to the adjacent dermis of the skin.
- The first pair of somites appears at the end of the third week a short distance caudal to the site at which the otic placode forms.



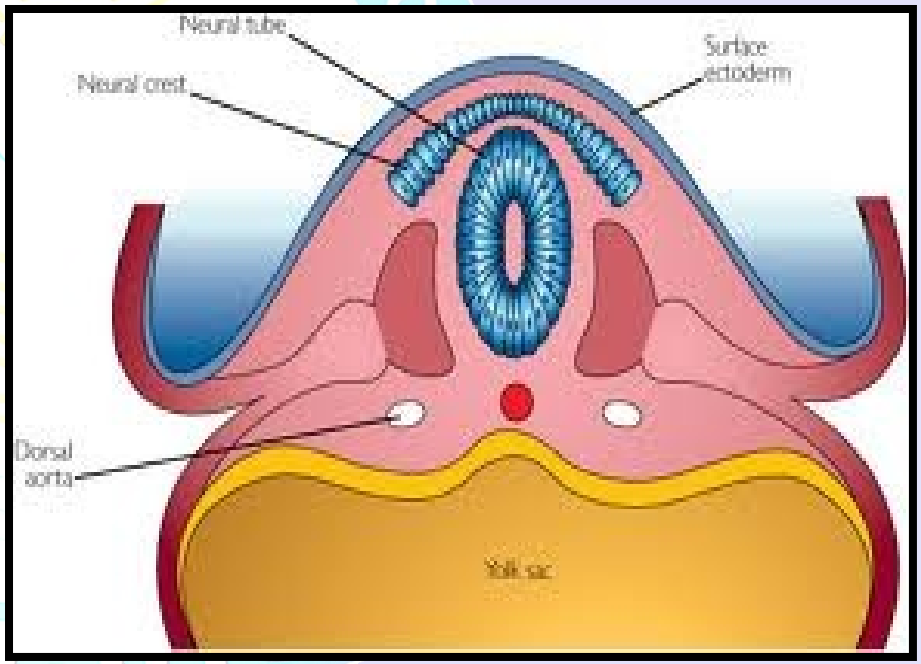
DEVELOPMENT OF THE INTRAEMBRYONIC COELOM

- The primordium of the intraembryonic coelom (embryonic body cavity) appears as isolated *coelomic spaces* in the lateral mesoderm and cardiogenic (heart-forming) mesoderm.
- These spaces soon coalesce to form a single horseshoe-shaped cavity, the **intraembryonic coelom**, which divides the lateral mesoderm into two layers:
- A somatic or *parietal layer* of lateral mesoderm located beneath the ectodermal epithelium and continuous with the extraembryonic mesoderm covering the amnion



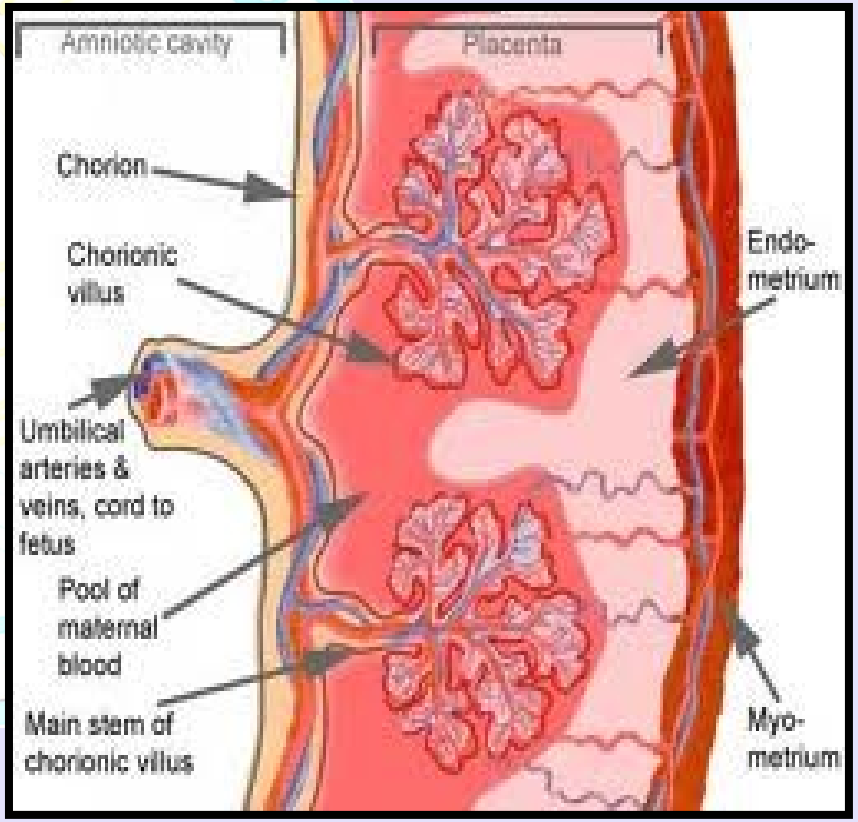
SPLANCHNIC OR *VISCERAL LAYER*

- A splanchnic or *visceral layer* of lateral mesoderm located adjacent to the endoderm and continuous with the extraembryonic mesoderm covering the umbilical vesicle



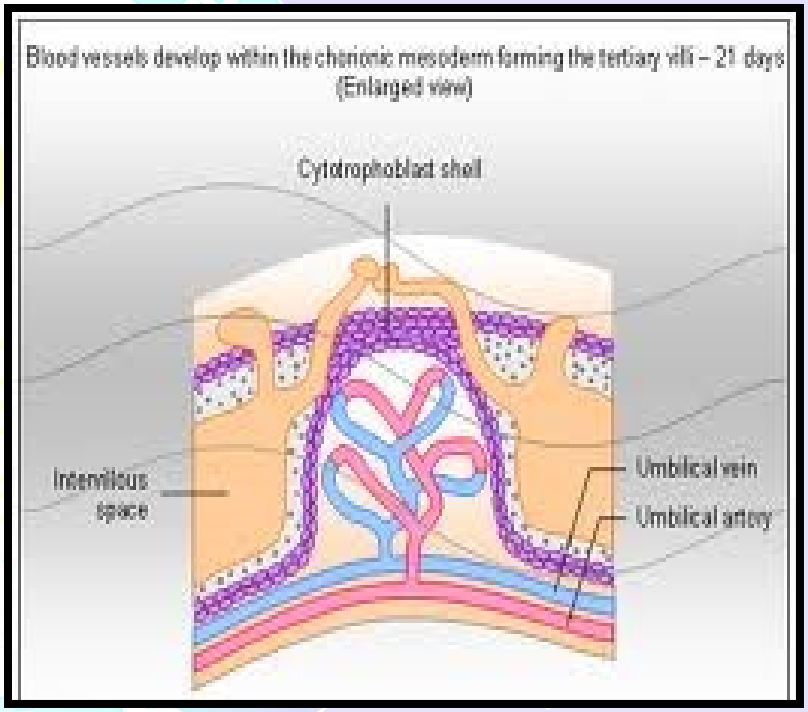
DEVELOPMENT OF CHORIONIC VILLI

- Shortly after **primary chorionic villi** appear at the end of the second week, they begin to branch.
- Early in the third week, mesenchyme grows into these primary villi, forming a core of mesenchymal tissue.
- The villi at this stage- **secondary chorionic villi**- cover the entire surface of the chorionic sac.
- Some mesenchymal cells in the villi soon differentiate into capillaries and blood cells



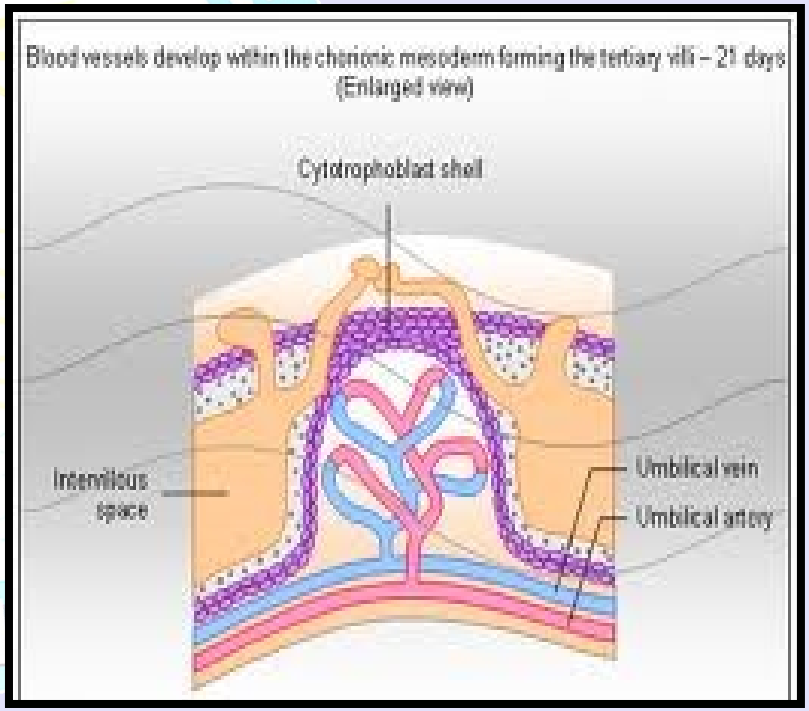
DEVELOPMENT OF CHORIONIC VILLI

- They are called **tertiary chorionic villi** when blood vessels are visible in them.
- The capillaries in the chorionic villi fuse to form **arteriocapillary networks**, which soon become connected with the embryonic heart through vessels that differentiate in the mesenchyme of the chorion and connecting stalk.



DEVELOPMENT OF CHORIONIC VILLI

- By the end of the third week, embryonic blood begins to flow slowly through the capillaries in the chorionic villi.
- Oxygen and nutrients in the maternal blood in the intervillous space diffuse through the walls of the villi and enter the embryo's blood.



REFERENCES

- **Keith L. Moore Developing Human**

  8th Edition

  Chapter-4

  pages 65-70



THANK YOU